Vidyavardhini's College of Engineering & Technology

Department of Computer Science & Engineering (Data Science

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Div : SE2

Roll No: 19

Branch : CSE-DS

Experiment No : 5

Experiment Title : Develop a web applica on to fetch real- me weather data for a city. Use AJAX & a public weather API to retrieve informa on. Display temperature, humidity, and weather condi ons dynamically.

AIM : To implement an interac ve weather monitoring system using AJAX requests and the OpenWeatherMap API, enabling users to fetch and view live weather parameters for any city.

CODE :

Index.html :

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    < tle>Weather App</ tle>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<div class="container">
    <h1>🌥 Real-Time Weather App</h1>

    <div class="input-box">
        <input type="text" id="city" placeholder="Enter city name...">
        <bu on onclick="getWeather()">Get Weather</bu on>
    </div>

    <div id="result" class="result-box"></div>
</div>

<script src="script.js"></script>
</body>
</html>
```

Style.css :

```
/* Background */ body {
margin: 0;     font-family:
'Poppins', sans-serif;
   background:
url('h ps://upload.wikimedia.org/wikipedia/commons/7/73/Cloudy_sky_%2826171935906%29
.jpg')  no-repeat  center  center/cover;
height: 100vh;     display: flex;     jus
fy-content:  center;         align-items:
center;
}

/* Main Glassmorphism Container */
.container {     background: rgba(255, 255,
255, 0.18);     padding: 40px 35px;     width:
420px;     border-radius: 20px;     text-align:
center;     backdrop-filter: blur(14px);     box-
shadow: 0 10px 35px rgba(0, 0, 0, 0.25);
color: #222; /* DARK TEXT */
}

/*  Heading  */  h1  {        font-size:
32px;         margin-bo  om:  25px;
color: #222; /* Dark Heading */
}
```

```
/* Input Field */ input {       width: 80%;
padding: 12px;    border: none;    outline:
none;    border-radius: 10px;    font-size:
17px;    background: rgba(255, 255, 255,
0.75);      color: #222; /* Dark Input Text
*/    margin-bo om: 20px;
}

/* Bu on */ bu
on {
   padding: 12px 25px;
font-size: 16px;
border: none;    border-
radius: 10px;
background: #006eff;
color: white;    cursor:
pointer;    transi on:
0.3s;
}

bu on:hover {
background: #0052cc;
}

/* Weather Result Box */ #result {
margin-top: 25px;    font-size: 19px;
```

```
padding: 20px;     border-radius: 15px;
background: rgba(255, 255, 255, 0.25);
backdrop-filter: blur(8px);     color:
#222; /* Dark Text */     line-height: 1.6;
}
```

Script.js :

```
func    on    getWeather()    {                    const    city    =
document.getElementById("city").value;     const apiKey =
"ee115b7622710cfeece37c786a8915d5";

   if (city === "") {         alert("Please
enter a city name!");         return;
   }

   const url =
`h ps://api.openweathermap.org/data/2.5/weather?q=${city}&appid=${apiKey}&units=metric`
;

   // AJAX request     const xhr = new
XMLH pRequest();
xhr.open("GET", url, true);
```

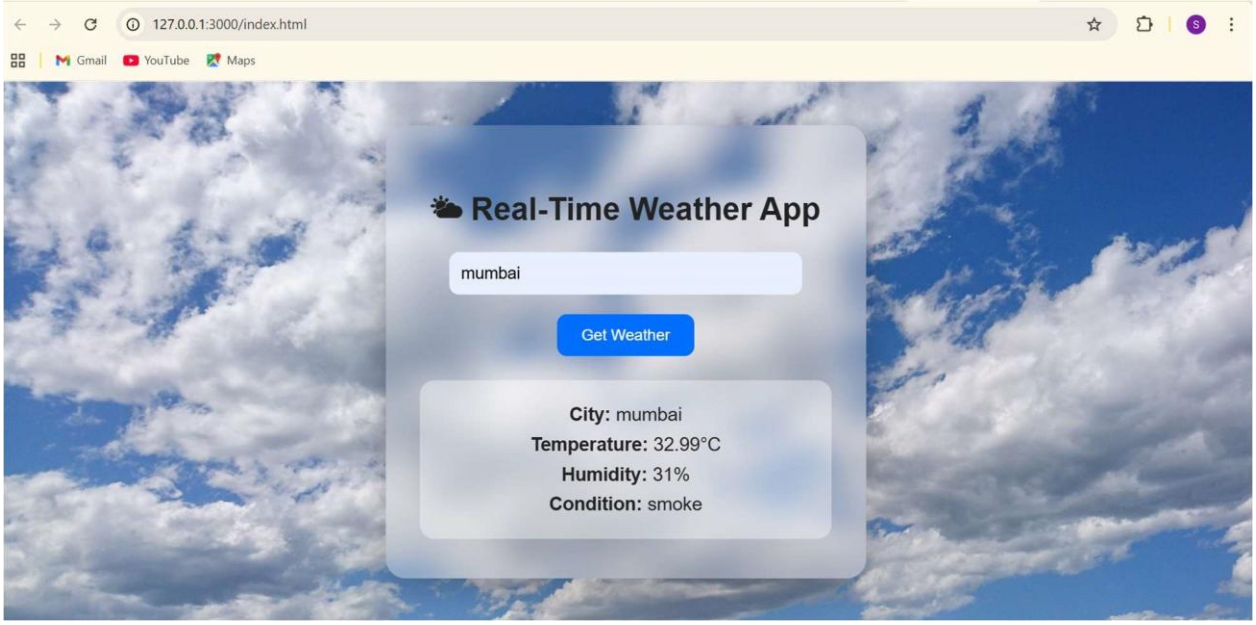
```
   xhr.onload = func on () {        if (this.status ===
200)   {                                const   data   =
JSON.parse(this.responseText);


          const temp = data.main.temp;                const
humidity = data.main.humidity;           const condi on
=                  data.weather[0].descrip                  on;
document.getElementById("result").innerHTML = `

             <strong>City:</strong> ${city}<br>
             <strong>Temperature:</strong> ${temp}°C <br>
             <strong>Humidity:</strong> ${humidity}% <br>
             <strong>Condi on:</strong> ${condi on}
          `;
}
else {
          document.getElementById("result").innerHTML = "□ City not found!";
       }
   };


   xhr.send();
}
```

OUTPUT:


127.0.0.1:3000/index.html
Gmail
YouTube
Maps
Real-Time Weather App
mumbai
Get Weather
City: mumbai
Temperature: 32.99°C
Humidity: 31%
Condition: smoke